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*Academy of Sciences of the Ukrainian SSR, Institute for Theoretical Physics, Kiev***Solitary Excitons in One-Dimensional Molecular Chains**

By

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The change in time of the state of intramolecular excitation in an one-dimensional molecular chain is investigated, where its deformability is taken into account. It is shown that the excitation which is accompanied by a local deformation of the chain, can move uniformly along the chain, the size of the region under excitation remaining constant. These excitations in one-dimensional molecular chains can be called *particle-like excitons* or *solitary excitons*.

Es werden die zeitlichen Änderungen des intramolekularen Anregungszustandes in einer ein-dimensionalen molekularen Kette untersucht, wobei deren Deformierbarkeit berücksichtigt wird. Es wird gezeigt, daß die Anregung der Kette, die durch eine lokale Deformation begleitet ist, sich gleichförmig längs der Kette bewegen kann, wobei die Größe des angeregten Bereichs konstant bleibt. Diese Anregungen in ein-dimensionalen Molekülketten können *partikel-ähnliche Exzitonen* oder *solitäre Exzitonen* genannt werden.

1. One-Dimensional Molecular Chain

The stationary states of the electron excitation taking crystal lattice deformation into account were considered in [1, 2]. The change of such a state in time is of certain interest. The present paper is devoted to this problem.

Consider a one-dimensional infinite molecular chain. The Hamiltonian of the electron excitation in the Heitler-London approximation has the form (see [3]):

$$H = \sum_n [\varepsilon + D(n)] B_n^+ B_n + \sum'_{l,n} M_{ln} (r_l - r_n) B_n^+ B_n, \quad (1)$$

where ε is the excitation energy of an isolated molecule;

$$D(n) = \sum_{l(\neq n)} D_{ln} (r_l - r_n),$$

$D_{ln} (r_l - r_n)$ is the change in the interaction energy between molecules l and n , when the molecule n is excited; $M_{ln} (r_l - r_n)$ are the matrix elements of the resonance interaction; r_n is the coordinate along the chain of the molecule n ; B_n^+ , B are the creation and annihilation operators of the electronic excitation over the molecule n .

The kinetic energy T of the molecules of the chain is

$$T = \frac{m}{2} \sum_n \left(\frac{\partial r_n}{\partial t} \right)^2, \quad (2)$$

where m is the mass of the molecule.

In the nearest-neighbour approximation the potential energy of the chain is

$$U = \sum_n u(R_n), \quad (3)$$

where $R_n = r_{n+1} - r_n > 0$ are the intermolecular distances;

$$u(R_n) = \frac{C}{R_n^s} - \frac{A}{R_n^q},$$

$C, A, s, q > 0$ are the constants defining the interaction of unexcited molecules; $s > q$.

In the ground state $|0\rangle$ (without intramolecular excitations) the chain possesses periodicity and the intermolecular distances obtained from the condition of a minimum of potential energy (3) are equal to

$$R = \left(\frac{sC}{qA} \right)^{1/(s-q)}. \quad (4)$$

In the state $|\psi\rangle$ when the intramolecular excitation is present, the intermolecular distances R_n are different from R and the potential energy of the chain changes by the value ΔU , as compared to the energy of the chain in the ground state. In the harmonic approximation

$$\Delta U = \sum_n [u(R_n) - u(R)] \approx w \sum_n \varrho_n^2, \quad (5)$$

where

$$R_n = R - \varrho_n, \quad |\varrho_n| < R, \quad (6)$$

$$w = \frac{1}{2} \left[\frac{\partial^2 u}{\partial R_n^2} \right]_{R_n=R}. \quad (7)$$

Taking into account (6), the formula (2) for the kinetic energy of the molecules can be transformed to

$$T = \sum_n \left\{ \frac{m}{2} \left[\sum_i \frac{\partial \varrho_{i+n}}{\partial t} \right]^2 \right\}. \quad (8)$$

Then the total energy E of the deformation of the chain is written as

$$E = T + \Delta U = \sum_n \left\{ \frac{m}{2} \left[\sum_{i=n}^{\infty} \frac{\partial \varrho_i}{\partial t} \right]^2 + w \varrho_n^2 \right\}. \quad (9)$$

2. The Nearest-Neighbour Approximation

In the nearest-neighbour approximation the operator (1) takes on the form

$$H = \sum_n \{ [\varepsilon + D(R_n)] B_n^+ B_n + M(R_n) [B_n^+ B_{n+1} + B_{n+1}^+ B_n] \}. \quad (10)$$

In (10) we adopt

$$D_{n-1,n}(R_{n-1}) + D_{n+1,n}(R_n) \approx D(R_n), \\ M_{n,n+1}(R_n) \approx M_{n+1,n}(R_n) = M(R_n).$$

Let

$$D(R_n) = -\frac{\alpha}{R_n^p}, \quad (11)$$

where $\alpha, p > 0$ are the parameters of theory.

In the case of a dipole-dipole resonance excitation (and only this excitation is considered below)

$$M(R_n) = \frac{b}{R_n^3}. \quad (12)$$

We assume that the dipole moment of transition, $\mathbf{d}$ is oriented along the chain. Then in (12) $b = -2\bar{d}^2$. Suppose that the following inequality is valid for distances close to R :

$$D > M, \quad (13)$$

where $D \equiv -D(R) > 0$; $M \equiv -M(R) > 0$. Using (6), (11), and (12), $D(R_n)$ and $M(R_n)$ in (10) can be expanded near the point R . In $D(R_n)$ we keep only the term linear in ϱ_n , and in $M(R_n)$ the zero term, i.e. we set $D(R_n) \approx \approx -(1 + p\varrho_n/R)D$, $M(R_n) \approx -M$. Then (10) takes the form

$$H = \sum_n \{[\varepsilon + (1 + p\varrho_n/R)D] B_n^+ B_n - M [B_n^+ B_{n+1} + B_{n+1}^+ B_n]\}. \quad (14)$$

The operator $\mathcal{H}$ of the excitation energy of the chain with its deformation energy taken into account reads

$$\mathcal{H} = H + E. \quad (15)$$

The solutions of the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \mathcal{H} |\Psi(t)\rangle \quad (16)$$

are looked for in the form

$$|\Psi(t)\rangle = \sum_n a_n(t) B_n^+ |0\rangle \quad (17)$$

provided

$$\langle \Psi(t) | \Psi \rangle = \sum_n |a_n(t)|^2 = 1. \quad (18)$$

The operator (15) is the function of a set of the values of ϱ_n . Moreover, these values are dependent on the state $|\Psi(t)\rangle$.

In the region of the chain where the density of the probability of finding the excitation $|a_n(t)|^2$ is great, one should expect greater deformation of the chain which is defined by the values of ϱ_n . From the condition of a minimum of the total excitation energy $\langle \Psi(t) | \mathcal{H} | \Psi(t) \rangle$ at every fixed moment of time t , and the fact that $\partial \varrho_n / \partial t$ is independent of ϱ_n , we find

$$\varrho_n = \frac{pD}{2wR} |a_n(t)|^2. \quad (19)$$

Substituting (19) into (14), we obtain

$$H = \sum_n \{[\varepsilon - D] B_n^+ B_n - M [B_n^+ B_{n+1} + B_{n+1}^+ B_n] - G |a_n(t)|^2 B_n^+ B_n\}, \quad (20)$$

where $G = p^2 D^2 / 2wR^2$ is the parameter of nonlinearity. Further, according to (9) and (19)

$$E = \frac{1}{2} \sum_n \left\{ F \left[\sum_{l=n} \frac{\partial}{\partial t} |a_l(t)|^2 \right]^2 + G |a_n(t)|^4 \right\}, \quad (21)$$

where

$$F = \frac{mG}{2w}.$$

Now the equation (16) takes the form

$$\sum_n \left\{ i\hbar \frac{\partial a_n(t)}{\partial t} + e a_n(t) - M [a_{n+1}(t) + a_{n-1}(t)] - G |a_n(t)|^2 a_n(t) \right\} B_n^+ |0\rangle = 0, \quad (22)$$

where

$$e = \varepsilon - D + E.$$

Using (22) and the fact that the system $B_n^+|0\rangle$ is complete and orthogonal, we obtain the equation for $a_n(t)$

$$i\hbar \frac{\partial a_n(t)}{\partial t} + M[a_{n+1}(t) + a_{n-1}(t)] - e a_n(t) + G|a_n(t)|^2 a_n(t) = 0. \quad (23)$$

In (23) we go over to a long-wave approximation and replace the differences by the derivatives. Thus

$$a_n(t) \rightarrow \varphi(\xi, t), \quad \varphi(n, t) = a_n(t), \quad (24)$$

$$\varphi(\xi + 1, t) + \varphi(\xi - 1, t) \approx 2\varphi(\xi, t) + \frac{\partial^2 \varphi(\xi, t)}{\partial \xi^2}, \quad (25)$$

where $\xi = r/R$ is the dimensionless coordinate. Then the differential-difference equation (23) transforms to

$$i\hbar \frac{\partial \varphi}{\partial t} + M \frac{\partial^2 \varphi}{\partial \xi^2} - \lambda \varphi + G|\varphi|^2 \varphi = 0, \quad (26)$$

where

$$\lambda \equiv \varepsilon - D - 2M + \frac{1}{2} \int_{-\infty}^{\infty} d\xi \left\{ F \left[\int_{\xi}^{\infty} d\eta \frac{\partial}{\partial t} |\varphi(\eta, t)|^2 \right]^2 + G|\varphi(\xi, t)|^2 \right\}, \quad (27)$$

3. The Particle-Like Excitation

The normalized solution of the nonlinear equation (26) has the form of a solitary wave

$$\varphi_v(\xi, t) = \frac{\sqrt{\frac{\mu}{2}} \exp \left\{ i \left[\frac{\hbar v}{2M} (\xi - \xi_0) - \Omega t \right] \right\}}{\cosh [\mu (\xi - \xi_0 - vt)]}, \quad (28)$$

where

$$\mu = \frac{G}{4M},$$

$$\hbar \Omega = \varepsilon - D - 2M - \frac{G^2}{48M} + \frac{m^* R^2 v^2}{2}, \quad (29)$$

$$m^* = m_0^* + m_d^* = \frac{\hbar^2 + (GF/6)}{2MR^2}; \quad (29a)$$

$m_0^* = \hbar^2/2MR^2$ is the effective mass of the excitation in the chain with rigidly fixed molecules; $m_d^* = mG^2/24MwR^2$ is the addition to the effective mass which is caused by a local deformation.

According to (28), the excitation is localized in the neighbourhood $\xi = \xi_0 + vt$. Consequently, the parameter defines the velocity $V = vR$ of the excitation motion along the chain. To every value of v there corresponds an excitation energy of the chain, $\hbar \Omega$. The excitation localized near the point ξ_0 corresponds to the value $v = 0$. Its energy $\varepsilon - D - 2M - G^2/48M$ is less by the value $G^2/48M$ than the excitation energy of the chain with rigidly fixed molecules.

Using (19), we find that in the excitation region the equilibrium distances between molecules decrease by the value

$$\varrho(\xi, t) = \frac{\sqrt{\frac{G}{2}} \frac{\mu}{w}}{\cosh^2 [\mu (\xi - \xi_0 - v t)]}. \quad (30)$$

Thus, an appreciable change in the distances between the molecules, which is proportional to the parameter μ , spreads over the region

$$\Delta \xi = \frac{\pi}{\mu} = \frac{4 \pi M}{G} \quad (31)$$

around the point $\xi_0 + v t$. This region travels with constant speed along the chain. The average distance between the molecules in the excitation region is

$$\langle R_n \rangle \approx R - 0.3 \mu \sqrt{\frac{G}{2 w}}. \quad (32)$$

Therefore, when the inertia and deformability of a chain is taken into account, the motion of the excitation is accompanied by a local distortion of the chain and the velocity V of an excitation decreases, as the effective mass (29a) increases.

Unlike the states of motion of free particles described by wave packets, the excitation state corresponding to the function (28) does not "spread" in the course of time. These excitations in one-dimensional molecular chains can be called *particle-like excitons*, or solitary excitons.

The long-wave approximation used by us requires that the inequality $\Delta \xi > 1$, i.e. $\mu = G/4 M < \pi$ is satisfied. On the other hand, for the particle-like excitons the following inequality should be valid:

$$\Delta \xi \ll N \quad \text{i. e.} \quad \mu > \frac{\pi}{N},$$

where N is the total number of molecules in the chain; $N \gg 1$.

At the very small values of the parameter μ ($\mu \ll 1$) the expression (25) can be transformed to the form of a simple harmonic wave

$$\varphi_v(\xi, t) \approx \sqrt{\frac{\mu}{2}} \exp \left\{ i \left[\frac{\hbar v}{2 M} (\xi - \xi_0) - \Omega t \right] \right\}.$$

Consequently, in this limiting case the excitation is practically nonlocalized.

Excitations such as (28) can arise in the chain under the influence of an external perturbation $\Phi(\xi, t)$ which satisfies the inequality

$$\int \Phi(\xi, t) \varphi_v^*(\xi, t) d\xi \neq 0.$$

We have not taken into account the processes of relaxation of the arising excitation. Phenomenologically the relaxation processes can be taken into account by replacing the excitation energy (26) by the complex value $\hbar \Omega - i \hbar \gamma$. Then an exponentially decreasing factor $\exp(-\gamma t)$ appears in the function (28) determining the average life-time ($1/\gamma$) of the excitation.

References

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